

$$\therefore \quad \begin{aligned} f(1, -2) &= -10, f_x = 2xy, f_x(1, -2) = -4; f_y = x^2 + 3, f_y(1, -2) = 4; f_{xx} = 2y, \\ f_{xx}(1, -2) &= -4; f_{xy} = 2x, f_{xy}(1, -2) = 2; f_{yy} = 0, f_{yy}(1, -2) = 0; f_{xxx} = 0, f_{xxx}(1, -2) = 0; \\ f_{xxy} &= 2, f_{xxy}(1, -2) = 0, f_{yyy}(1, -2) = 0 \end{aligned}$$

All partial derivatives of higher order vanish.

Substituting these in (i), we get

$$\begin{aligned} x^2y + 3y - 2 &= -10 + [(x-1)(-4) + (y+2)4] + \frac{1}{2} [(x-1)^2(-4) + 2(x-1)(y+2)(2) \\ &\quad + (y+2)^2(0)] + \frac{1}{6} [(x-1)^3(0) + 3(x-1)^2(y+2)(2) + 3(x-1)(y+2)^2(0) + (y+2)^3(0)] \\ &= -10 - 4(x-1) + 4(y+2) - 2(x-1)^2 + 2(x-1)(y+2) + (x-1)^2(y+2). \end{aligned}$$

Example 5.34. Expand $f(x, y) = \tan^{-1}(y/x)$ in powers of $(x-1)$ and $(y-1)$ upto third-degree terms. Hence compute $f(1.1, 0.9)$ approximately. (V.T.U., 2010 ; J.N.T.U., 2006 ; U.P.T.U., 2006)

Solution. Here $a = 1, b = 1$ and $f(1, 1) = \tan^{-1}(1) = \pi/4$.

$$\begin{aligned} f_x &= \frac{-y}{x^2 + y^2} & f_x(1, 1) &= -\frac{1}{2}; & f_y &= \frac{x}{x^2 + y^2}, & f_y(1, 1) &= \frac{1}{2} \\ f_{xx} &= \frac{2xy}{(x^2 + y^2)^2}, & f_{xx}(1, 1) &= \frac{1}{2}; & f_{xy} &= \frac{y^2 - x^2}{(x^2 + y^2)^2}, & f_{xy}(1, 1) &= 0 \\ f_{yy} &= \frac{-2xy}{(x^2 + y^2)^2}, & f_{yy}(1, 1) &= -\frac{1}{2}; \\ f_{xxx} &= \frac{2y^3 - 6x^2y}{(x^2 + y^2)^3}, & f_{xxx}(1, 1) &= -\frac{1}{2}; & f_{xxy} &= \frac{2x^3 - 6xy^2}{(x^2 + y^2)^3}, & f_{xxy}(1, 1) &= -\frac{1}{2} \\ f_{xyy} &= \frac{6x^2y - 2y^3}{(x^2 + y^2)^3}, & f_{xyy}(1, 1) &= \frac{1}{2}; & f_{yyy} &= \frac{6xy^2 - 2x^3}{(x^2 + y^2)^3}, & f_{yyy}(1, 1) &= \frac{1}{2} \end{aligned}$$

Taylor's expansion of $f(x, y)$ in powers of $(x-1)$ and $(y-1)$ is given by

$$\begin{aligned} f(x, y) &= f(1, 1) + \frac{1}{1!} [(x-1)f_x(1, 1) + (y-1)f_y(1, 1)] + \frac{1}{2!} [(x-1)^2 f_{xx}(1, 1) + 2(x-1)(y-1) \\ &\quad f_{xy}(1, 1) + (y-1)^2 f_{yy}(1, 1)] + \frac{1}{3!} [(x-1)^3 f_{xxx}(1, 1) + 3(x-1)^2(y-1)f_{xxy}(1, 1) \\ &\quad + 3(x-1)(y-1)^2 f_{xyy}(1, 1) + (y-1)^3 f_{yyy}(1, 1)] + \dots \\ \therefore \tan^{-1}\left(\frac{y}{x}\right) &= \frac{\pi}{4} + \left\{ (x-1)\left(-\frac{1}{2}\right) + (y-1)\frac{1}{2} \right\} + \frac{1}{2!} \left\{ (x-1)^2 \frac{1}{2} + 2(x-1)(y-1)(0) + (y-1)^2 \left(-\frac{1}{2}\right) \right\} \\ &\quad + \frac{1}{3!} \left\{ (x-1)^3 \left(-\frac{1}{2}\right) + 3(x-1)^2(y-1)\left(-\frac{1}{2}\right) + 3(x-1)(y-1)^2 \frac{1}{2} + (y-1)^3 \frac{1}{2} \right\} + \dots \\ &= \frac{\pi}{4} - \frac{1}{2} [(x-1) - (y-1)] + \frac{1}{4} [(x-1)^2 - (y-1)^2] - \frac{1}{12} [(x-1)^3 + 3(x-1)^2(y-1) \\ &\quad - 3(x-1)(y-1)^2 - (y-1)^3] + \dots \end{aligned}$$

Putting $x = 1.1$ and $y = 0.9$, we get

$$\begin{aligned} f(1.1, 0.9) &= \frac{\pi}{4} - \frac{1}{2}(0.2) + \frac{1}{4}(0) - \frac{1}{12} [(0.1)^3 - 3(0.1)^2(-0.1) - 3(0.1)(-0.1)^2 - (-0.1)^3] \\ &= 0.7854 - 0.1000 + 0.0003 = 0.6857. \end{aligned}$$

5.10 (1) ERRORS AND APPROXIMATIONS

Let $f(x, y)$ be a continuous function of x and y . If δx and δy be the increments of x and y , then the new value of $f(x, y)$ will be $f(x + \delta x, y + \delta y)$. Hence

$$\delta f = f(x + \delta x, y + \delta y) - f(x, y).$$

Expanding $f(x + \delta x, y + \delta y)$ by Taylor's theorem and supposing $\delta x, \delta y$ to be so small that their products, squares and higher powers can be neglected, we get

$$\delta f = \frac{\partial f}{\partial x} \delta x + \frac{\partial f}{\partial y} \delta y, \text{ approximately.}$$

Similarly if f be a function of several variables $x, y, z, t, \dots$, then

$$\delta f = \frac{\partial f}{\partial x} \delta x + \frac{\partial f}{\partial y} \delta y + \frac{\partial f}{\partial z} \delta z + \frac{\partial f}{\partial t} \delta t + \dots \text{ approximately.}$$

These formulae are very useful in correcting the effect of small errors in measured quantities.

(2) Total Differential

If u is a function of two variables x and y , the *total differential* of u is defined as

$$du = \frac{\partial u}{\partial x} dx + \frac{\partial u}{\partial y} dy \quad \dots(1)$$

The differentials dx and dy are respectively the increments δx and δy in x and y . If x and y are not independent variables but functions of another variable t even then the formula (1) holds and we write $dx = \frac{dx}{dt} dt$ and $dy = \frac{dy}{dt} dt$. Similar definition can be given for a function of three or more variables.

Example 5.35. The diameter and altitude of a can in the shape of a right circular cylinder are measured as 4 cm and 6 cm respectively. The possible error in each measurement is 0.1 cm. Find approximately the maximum possible error in the values computed for the volume and the lateral surface.

Solution. Let x be the diameter and y the height of the can. Then its volume $V = \frac{\pi}{4} x^2 y$

$$\therefore \delta V = \frac{\partial V}{\partial x} \delta x + \frac{\partial V}{\partial y} \delta y = \frac{\pi}{4} (2xy \delta x + x^2 \delta y)$$

When $x = 4$ cm., $y = 6$ cm. and $\delta x = \delta y = 0.1$ cm.

$$\therefore \delta V = \frac{\pi}{4} (2 \times 4 \times 6 \times 0.1 + 4^2 \times 0.1) = 1.6\pi \text{ cm}^3$$

Also its lateral surface $S = \pi xy$

$$\therefore \delta S = \pi(y \delta x + x \delta y)$$

When $x = 4$ cm., $y = 6$ cm. and $\delta x = \delta y = 0.1$ cm., we have $\delta S = \pi(6 \times 0.1 + 4 \times 0.1) = \pi \text{ cm}^2$.

Example 5.36. The period of a simple pendulum is $T = 2\pi \sqrt{l/g}$, find the maximum error in T due to the possible error upto 1% in l and 2.5% in g . (U.P.T.U., 2004)

Solution. We have $T = 2\pi \sqrt{l/g}$

$$\text{or} \quad \log T = \log 2\pi + \frac{1}{2} \log l - \frac{1}{2} \log g$$

$$\therefore \frac{1}{T} \delta T = 0 + \frac{1}{2} \frac{1}{l} \delta l - \frac{1}{2} \frac{1}{g} \delta g$$

$$\frac{\delta T}{T} 100 = \frac{1}{2} \left(\frac{\delta l}{l} 100 - \frac{\delta g}{g} 100 \right) = \frac{1}{2} (1 \pm 2.5) = 1.75 \text{ or } -0.75$$

Thus the maximum error in $T = 1.75\%$

Example 5.37. A balloon is in the form of right circular cylinder of radius 1.5 m and length 4 m and is surmounted by hemispherical ends. If the radius is increased by 0.01 m and length by 0.05 m, find the percentage change in the volume of balloon. (U.P.T.U., 2005)

Solution. Let the volume of the balloon (Fig. 5.3) be V , so that

$$V = \pi r^2 h + \frac{2}{3} \pi r^3 + \frac{2}{3} \pi r^3 = \pi r^2 h + \frac{4}{3} \pi r^3$$

$$\therefore \delta V = 2\pi r \delta r h + \pi r^2 \delta h + \frac{4}{3} \pi 3r^2 \delta r$$

$$\frac{\delta V}{V} = \frac{\pi r [2h \delta r + r \delta h + 4r \delta r]}{\pi r^2 h + \frac{4}{3} \pi r^3}$$

$$= \frac{2(h + 2r) \delta r + r \delta h}{rh + \frac{4}{3} r^2} = \frac{2(4 + 3)(.01) + 1.5(.05)}{1.5 \times 4 + \frac{4}{3} (1.5)^2}$$

$$= \frac{0.14 + 0.075}{6 + 3} = \frac{0.215}{9}$$

Hence, the percentage change in $V = 100 \frac{\delta V}{V} = \frac{21.5}{9} = 2.39\%$

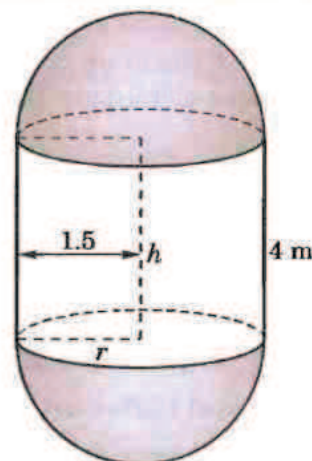


Fig. 5.3

Example 5.38. In estimating the cost of a pile of bricks measured as $2 \text{ m} \times 15 \text{ m} \times 1.2 \text{ m}$, the tape is stretched 1% beyond the standard length. If the count is 450 bricks to 1 cu. m. and bricks cost ₹ 530 per 1000, find the approximate error in the cost. (V.T.U., 2001)

Solution. Let x , y and z m be the length, breadth and height of the pile so that its volume $V = xyz$

$$\log V = \log x + \log y + \log z \therefore \frac{\delta V}{V} = \frac{\delta x}{x} + \frac{\delta y}{y} + \frac{\delta z}{z}$$

Since $V = 2 \times 15 \times 1.2 = 36 \text{ m}^3$, and $\frac{\delta x}{x} = \frac{\delta y}{y} = \frac{\delta z}{z} = \frac{1}{100}$

$$\therefore \delta V = 36 \left(\frac{3}{100} \right) = 1.08 \text{ m}^3.$$

Number of bricks in $\delta V = 1.08 \times 450 = 486$

Thus error in the cost = $486 \times \frac{530}{1000} = ₹ 257.58$ which is a loss to the brick seller.

Example 5.39. The height h and semi-vertical angle α of a cone are measured and from them A , the total area of the surface of the cone including the base is calculated. If h and α are in error by small quantities δh and $\delta \alpha$ respectively, find the corresponding error in the area. Show further that if $\alpha = \pi/6$, an error of + 1% in h will be approximately compensated by an error of - 0.33 degrees in α .

Solution. If r be the base radius and l the slant height of the cone, (Fig. 5.4), then total area

$$A = \text{area of base} + \text{area of curved surface}$$

$$= \pi r^2 + \pi r l = \pi r(r + l)$$

$$= \pi h \tan \alpha (h \tan \alpha + h \sec \alpha)$$

$$= \pi h^2 (\tan^2 \alpha + \tan \alpha \sec \alpha)$$

$$\therefore \delta A = \frac{\delta A}{\delta h} \delta h + \frac{\delta A}{\delta \alpha} \delta \alpha$$

$$= 2\pi h (\tan^2 \alpha + \tan \alpha \sec \alpha) \delta h$$

$$+ \pi h^2 (2 \tan \alpha \sec^2 \alpha + \sec^3 \alpha + \tan \alpha \sec \alpha \tan \alpha) \delta \alpha$$

which gives the error in the area A .

Putting $\delta h = h/100$ and $\alpha = \pi/6$, we get

$$\delta A = 2\pi h \left[\left(\frac{1}{\sqrt{3}} \right)^2 + \frac{1}{\sqrt{3}} \cdot \frac{2}{\sqrt{3}} \right] \frac{h}{100} + \pi h^2 \left[2 \cdot \frac{1}{\sqrt{3}} \cdot \frac{4}{3} + \frac{8}{3\sqrt{3}} + \frac{1}{\sqrt{3}} \cdot \frac{2}{\sqrt{3}} \cdot \frac{1}{\sqrt{3}} \right] \delta \alpha$$

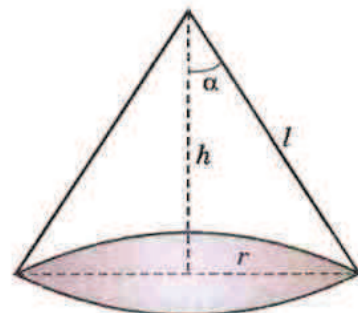


Fig. 5.4

$$= \frac{2\pi h^2}{100} + 2\sqrt{3}\pi h^2 \delta\alpha$$

The error in h will be compensated by the error in α , when

$$\delta A = 0 \text{ i.e., } \frac{2\pi h^2}{100} + 2\sqrt{3}\pi h^2 \delta\alpha = 0$$

or
$$\delta\alpha = -\frac{1}{100\sqrt{3}} \text{ radians} = -\frac{.01}{1.732} \times 57.3^\circ = -0.33^\circ.$$

Example 5.40. Show that the approximate change in the angle A of a triangle ABC due to small changes δa , δb , δc in the sides a , b , c respectively, is given by

$$\delta A = \frac{a}{2\Delta} (\delta a - \delta b \cos C - \delta c \cos B)$$

where Δ is the area of the triangle. Verify that $\delta A + \delta B + \delta C = 0$.

Solution. We know that $a^2 = b^2 + c^2 - 2bc \cos A$

so that
$$2a\delta a = 2b\delta b + 2c\delta c - 2(c\delta b \cos A - b\delta c \cos A + bc \sin A \delta A)$$

$\therefore bc \sin A \delta A = a\delta a - (b - c \cos A) \delta b - (c - b \cos A) \delta c$

or
$$2\Delta \delta A = a\delta a - (c \cos A + a \cos C - c \cos A) \delta b - (a \cos B + b \cos A - b \cos A) \delta c$$

$$[\because b = c \cos A + a \cos C \text{ etc. } \dots(i)]$$

$$= a\delta a - a \cos C \delta b - a \cos B \delta c$$

or
$$\delta A = \frac{a}{2\Delta} (\delta a - \delta b \cos C - \delta c \cos B)$$

By symmetry, we have

$$\delta B = \frac{b}{2\Delta} (\delta b - \delta c \cos A - \delta a \cos C)$$

$$\delta C = \frac{c}{2\Delta} (\delta c - \delta a \cos B - \delta b \cos A)$$

$\therefore \delta A + \delta B + \delta C = \frac{1}{2\Delta} (a - b \cos C - c \cos B) \delta a + (b - c \cos A - a \cos C) \delta b$

$$+ (c - a \cos B - b \cos A) \delta c$$

$$= \frac{1}{2\Delta} [(a - a) \delta a + (b - b) \delta b + (c - c) \delta c] = 0 \quad [\text{By (i)}]$$

Example 5.41. If the sides of a plane triangle ABC vary in such a way that its circumradius remains constant, prove that $\frac{da}{\cos A} + \frac{db}{\cos B} + \frac{dc}{\cos C} = 0$.

Solution. The circumradius R of ΔABC is given by

$$R = \frac{a}{2 \sin A} = \frac{b}{2 \sin B} = \frac{c}{2 \sin C}$$

$\therefore a = 2R \sin A \quad [\because R \text{ is constant}]$

Taking differentials, $da = 2R \cos A dA$ or $\frac{da}{\cos A} = 2R dA$

Similarly, $\frac{db}{\cos B} = 2R dB, \frac{dc}{\cos C} = 2R dC$

$\therefore \frac{da}{\cos A} + \frac{db}{\cos B} + \frac{dc}{\cos C} = 2R (dA + dB + dC)$

Now $A + B + C = \pi$, gives $dA + dB + dC = 0 \quad \dots(i)$

Thus $\frac{da}{\cos A} + \frac{db}{\cos B} + \frac{dc}{\cos C} = 0 \quad [\text{By (i)}]$

PROBLEMS 5.9

- Expand the following functions as far as terms of third degree :
 (i) $\sin x \cos y$ (V.T.U., 2009) (ii) $e^x \sin y$ at $(-1, \pi/4)$ (Anna, 2009)
 (iii) $xy^2 + \cos xy$ about $(1, \pi/2)$. (Hissar, 2005 S ; V.T.U., 2003)
- Expand $f(x, y) = x^y$ in powers of $(x - 1)$ and $(y - 1)$. (U.T.U., 2009)
- If $f(x, y) = \tan^{-1} xy$, compute $f(0.9, -1.2)$ approximately.
- If the kinetic energy $k = mv^2/2g$, find approximately the change in the kinetic energy as w changes from 49 to 49.5 and v changes from 1600 to 1590. (V.T.U., 2006)
- Find the possible percentage error in computing the resistance r from the formula $1/r = 1/r_1 + 1/r_2$, if r_1, r_2 are both in error by 2%.
- The voltage V across a resistor is measured with an error h , and the resistance R is measured with an error k . Show that the error in calculating the power $W(V, R) = V^2/R$ generated in the resistor, is $VR^{-2}(2Rh - Vk)$. (V.T.U., 2009)
- Find the percentage error in the area of an ellipse if one per cent error is made in measuring the major and minor axes. (V.T.U., 2011)
- The time of oscillation of a simple pendulum is given by the equation $T = 2\pi\sqrt{l/g}$. In an experiment carried out to find the value of g , errors of 1.5% and 0.5% are possible in the values of l and T respectively. Show that the error in the calculated value of g is 0.5%. (Cochin, 2005)
- If $pv^2 = k$ and the relative errors in p and v are respectively 0.05 and 0.025, show that the error in k is 10%. (Mysore, 1999)
- If the H.P. required to propel a steamer varies as the cube of the velocity and square of the length. Prove that a 3% increase in velocity and 4% increase in length will require an increase of about 17% in H.P.
- The range R of a projectile which starts with a velocity v at an elevation α is given by $R = (v^2 \sin 2\alpha)/g$. Find the percentage error in R due to an error of 1% in v and an error of $\frac{1}{2}\%$ in α . (Kurukshetra, 2009)
- In estimating the cost of a pile of bricks measured as $6 \text{ m} \times 50 \text{ m} \times 4 \text{ m}$, the tape is stretched 1% beyond the standard length. If the count is 12 bricks in 1 m^3 and bricks cost ₹ 100 per 1000, find the approximate error in the cost. (U.T.U., 2010 ; U.P.T.U., 2005)
- The deflection at the centre of a rod of length l and diameter d supported at its ends, loaded at the centre with a weight w varies as wl^3d^{-4} . What is the increase in the deflection corresponding to $p\%$ increase in w , $q\%$ decrease in l and $r\%$ increase in d ?
- The work that must be done to propel a ship of displacement D for a distance s in time t is proportional to $(s^2 D^{2/3} / t^2)$. Find approximately the increase of work necessary when the displacement is increased by 1%, the time is diminished by 1% and the distance diminished by 2%.
- The indicated horse power l of an engine is calculated from the formula $l = PLAN/33,000$, where $A = \pi d^2/4$. Assuming that error of r per cent may have been made in measuring P, L, N and d , find the greatest possible error in l .
- The torsional rigidity of a length of wire is obtained from the formula $N = 8\pi Il/t^2 r^4$. If l is decreased by 2%, r is increased by 2%, t is increased by 1.5%, show that the value of N is diminished by 13% approximately. (V.T.U., 2003)
- If $x^2 + y^2 + z^2 - 2xyz = 1$, show that $\frac{dx}{\sqrt{1-x^2}} + \frac{dy}{\sqrt{1-y^2}} + \frac{dz}{\sqrt{1-z^2}} = 0$.
 [Hint. $2(x - yz) dx + 2(y - zx) dy + 2(z - xy) dz = 0$. Also $(x - yz)^2 = (1 - y^2)(1 - z^2)$, ...]

5.11 (1) MAXIMA AND MINIMA OF FUNCTIONS OF TWO VARIABLES

Def. A function $f(x, y)$ is said to have a **maximum** or **minimum** at $x = a, y = b$, according as

$$f(a + h, b + k) < \text{or} > f(a, b),$$

for all positive or negative small values of h and k .

In other words, if $\Delta = f(a + h, b + k) - f(a, b)$, is of the same sign for all small values of h, k , and if this sign is negative, then $f(a, b)$ is a maximum. If this sign is positive, $f(a, b)$ is a minimum.